

Some Basic Statistical Tools

Outline for Today

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1. Note on methods of learning
2. Convergence for sequences of random variables
3. Break
4. Concentration inequalities
5. Proof of the weak law of large numbers
6. (If time) Proof techniques

Methods of Learning

(Disclaimer: I am not a trained epistemologist)

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Testimony

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1. Read or hear a piece of knowledge
2. Decide whether to believe it (e.g. based on authority of speaker)
3. Learn by believing it

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Perception (Scientific Method)

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2. Formulate a hypothesis
3. Test the hypothesis with an experiment
4. Learn by analyzing the results of the experiment

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Most RL research fits into the scientific method.

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Reason (Mathematical Proof)

Most RL research fits into the scientific method.

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Perception (Scientific Method)

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Reason

(Mathematical Proof)

1. Formulate a question
2. Create a conjecture that answers the question
 1. *Define* all concepts
 2. State as a theorem
3. Prove or disprove the conjecture
4. Learn by considering the theorem/false conjecture and the proof

Easiest
←
Least
Reliable

Most Resources

More Reliable

Most Difficult →

: Most Reliable
Most General

Most RL research fits into the scientific method.

Many results are
not practical

Review



Review

Given a **probability space** $(\Omega, \mathcal{F}, P)$, and a **measurable space** $(E, \mathcal{E})$, an E -valued **random variable** is a **measurable function** $X : \Omega \rightarrow E$.



Review

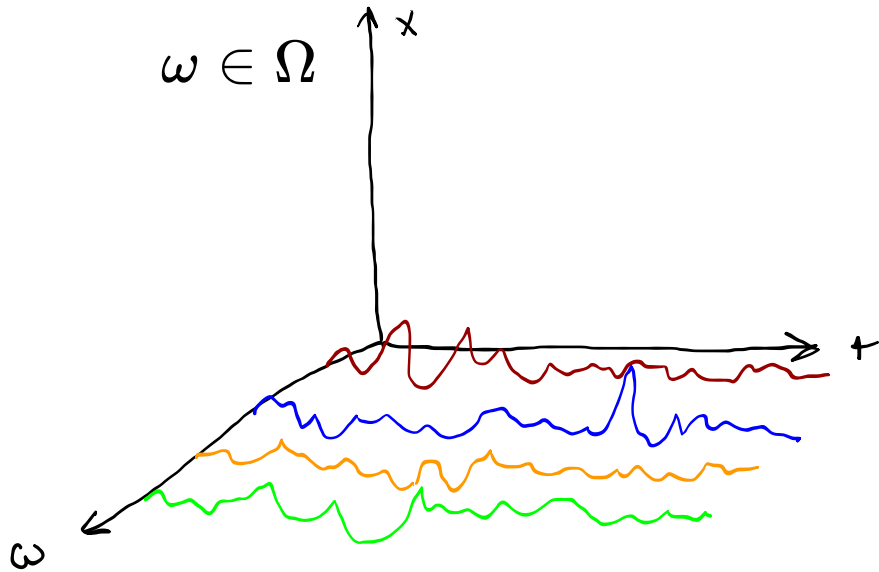
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$$\omega \in \Omega$$



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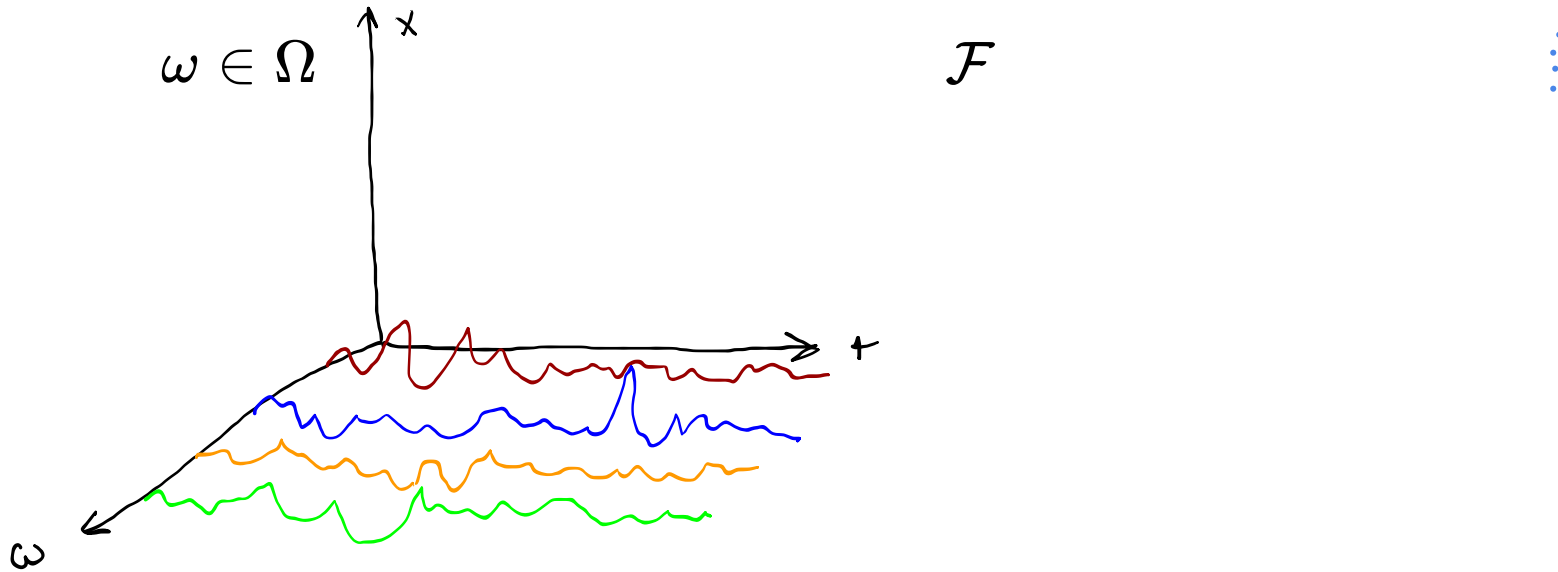
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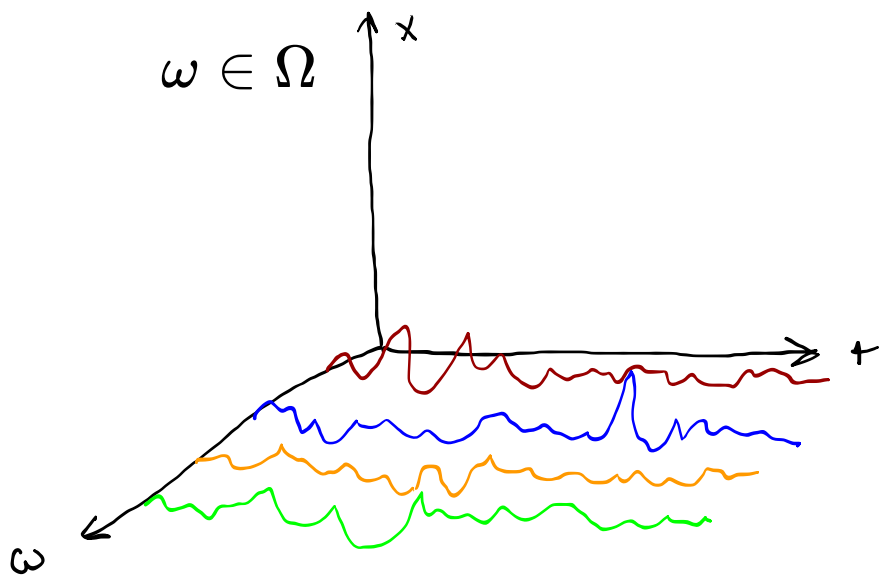
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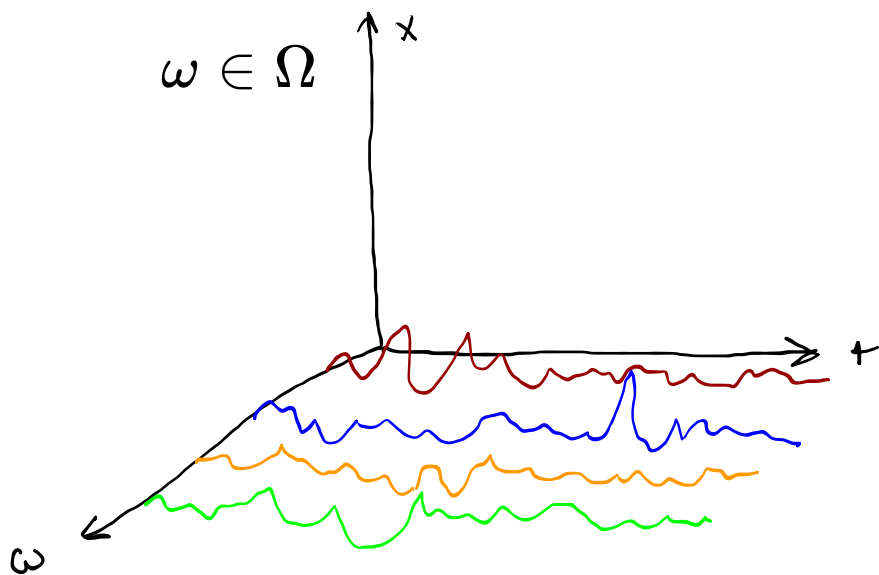
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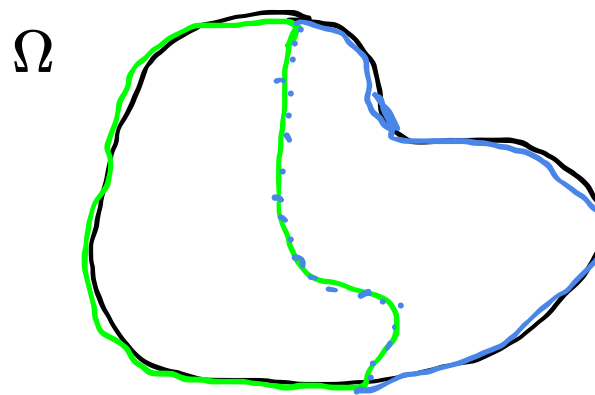
$$\mathcal{F} = \sigma(\{\text{green curve}\}) = \{\Omega, \text{green curve}, \text{blue curve}, \emptyset\}$$

Review

Given a **probability space** $(\Omega, \mathcal{F}, P)$, and a **measurable space** $(E, \mathcal{E})$, an E -valued **random variable** is a **measurable function** $X : \Omega \rightarrow E$.



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Review: Equality of Random Variables

Sure
"Equal"

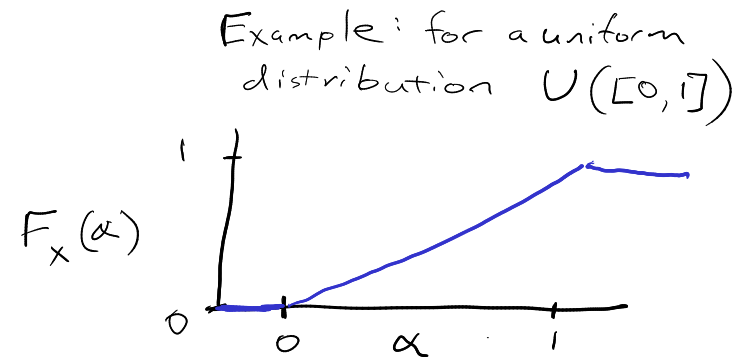
$$X = Y \iff X(\omega) = Y(\omega) \quad \forall \omega \in \Omega$$

Weak:
"In Law"
"In Distribution"

Cumulative distribution function

$$F_X(\alpha) \equiv P(X \leq \alpha)$$

$$X \stackrel{D}{=} Y \iff F_X(\alpha) = F_Y(\alpha) \quad \forall \alpha \in E$$



Almost Sure

"Equal almost everywhere"
"with probability 1"

$$X \stackrel{a.s.}{=} Y \iff \begin{aligned} &P(\{\omega : X(\omega) = Y(\omega)\}) = 1 \\ &P(\{\omega : X(\omega) \neq Y(\omega)\}) = 0 \end{aligned}$$

$$\Omega = \{-1, 1\}$$

$$P(-1) = 0.5$$

$$X(\omega) = \omega$$

$$Y(\omega) = -\omega$$

$$X \neq Y$$

$$X \stackrel{a.s.}{=} Y$$

$$X \stackrel{D}{=} Y$$

Review: Equality of Random Variables

Example:

$$E = \Omega = (0,1)$$

$$\mathcal{E} = \mathcal{F} = \mathcal{B} \text{ (Borel)} \\ \sigma(\text{all open sets})$$

$$P((a,b)) = b-a \\ \text{(uniform)}$$

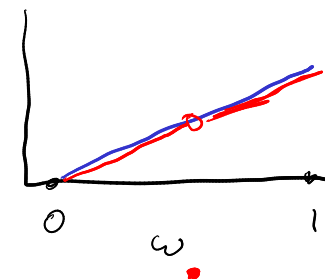
Property of probability measures

$$P\left(\bigcup_{i \in C} A^i\right) = \sum_{i \in C} P(A^i)$$

where A^i are disjoint
and C is countable

$$X(\omega) = \omega$$

$$Y(\omega) = \begin{cases} -1 & \text{if } \omega = 0.5 \\ X(\omega) & \text{o.w.} \end{cases}$$



For any $t \in (0,1)$ e.g. 0.5

$$\begin{aligned} P(\{t\}) &= 1 - P((0,t) \cup (t,1)) \\ &= 1 - (P((0,t)) + P((t,1))) \\ &= 1 - (t + (1-t)) \\ &= 0 \end{aligned}$$

$$P(\{0.5\}) = 0$$

$$X \stackrel{\text{a.s.}}{=} Y \quad \text{but} \quad X \neq Y$$

Lisa's question:

Is $\{t_1, t_2\}$ a zero measure set?

$$\text{i.e. } P(\{t_1, t_2\}) \stackrel{?}{=} 0$$

$$\begin{aligned} P(\{t_1, t_2\}) &= P(\{t_1\}) + P(\{t_2\}) \\ &= 0 \end{aligned}$$

$$\begin{aligned} P((0,t_1) \cup (t_1, t_2) \cup (t_2, 1)) \\ = 0 \end{aligned}$$

~~For~~

Review: Convergence

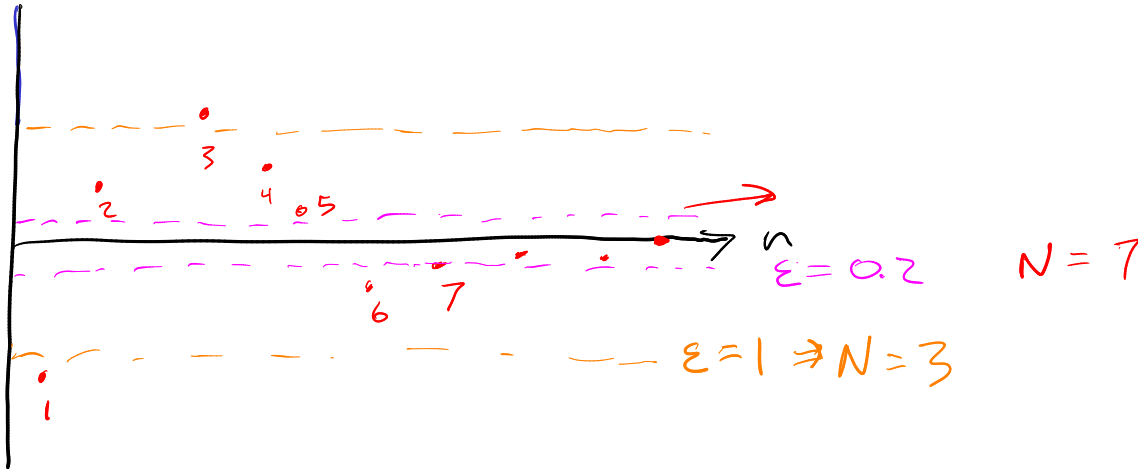
For a (deterministic) sequence $\{x_n\}$, we say

$$\lim_{n \rightarrow \infty} x_n = x$$

or

$$x_n \rightarrow x$$

if, for every $\epsilon > 0$, there exists an N such that $|x_n - x| < \epsilon$ for all $n > N$.



Random Variables: Sure Convergence

Sequence of random variables $\{X_n\} \xrightarrow{?} X$

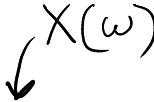
$$X_n(\omega) \rightarrow X(\omega) \quad \forall \omega \in \underline{\Omega}$$

Random Variables: Almost Sure Convergence

Random Variables:

Almost Sure Convergence

$X_n \xrightarrow{a.s.} X$ if and only if $P(\{\omega : X_n(\omega) \rightarrow X\}) = 1$, that is, X_n converges to X except possibly on a zero-measure set.



Random Variables:

Almost Sure Convergence

$X_n \xrightarrow{a.s.} X$ if and only if $P(\{\omega : X_n(\omega) \rightarrow X\}) = 1$, that is, X_n converges to X except possibly on a zero-measure set.

Does sure convergence imply almost sure convergence?

Yes

$$P(\Omega) = 1$$

Random Variables: Convergence in Probability

Random Variables:

Convergence in Probability

$X_n \xrightarrow{p} X$ if $P(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) \rightarrow 0$ for any fixed $\epsilon > 0$.

Choose ϵ

Random Variables:

Convergence in Probability

$X_n \rightarrow_p X$ if $P(\{\omega : |X_n(\omega) - X(\omega)| > \epsilon\}) \rightarrow 0$ for any fixed $\epsilon > 0$.

Does $X_n \xrightarrow{a.s.} X$ imply $X_n \rightarrow_p X$?

Yes

Random Variables: Convergence in Probability

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Random Variables: Convergence in Probability

Does $X_n \rightarrow_p X$ imply $X_n \xrightarrow{a.s.} X$?

No.

Random Variables:

Convergence in Probability

Does $X_n \rightarrow_p X$ imply $X_n \xrightarrow{a.s.} X$?

No.

PROOF. Consider the probability space $\Omega = (0, 1)$, with Borel σ -field and the Uniform probability measure U of Example 1.1.11. Suffices to construct an example of $X_n \rightarrow_p 0$ such that fixing each $\omega \in (0, 1)$, we have that $X_n(\omega) = 1$ for infinitely many values of n . For example, this is the case when $X_n(\omega) = \mathbf{1}_{[t_n, t_n + s_n]}(\omega)$ with $s_n \downarrow 0$ as $n \rightarrow \infty$ slowly enough and $t_n \in [0, 1 - s_n]$ are such that any $\omega \in [0, 1]$ is in infinitely many intervals $[t_n, t_n + s_n]$. The latter property applies if $t_n = (i - 1)/k$ and $s_n = 1/k$ when $n = k(k - 1)/2 + i$, $i = 1, 2, \dots, k$ and $k = 1, 2, \dots$ (plot the intervals $[t_n, t_n + s_n]$ to convince yourself). ■

Random Variables: Convergence in Probability

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But there exists a subsequence n_k such that $X_{n_k} \xrightarrow{a.s.} X$.

Random Variables: Weak Convergence

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$X_n \xrightarrow{D} X$ if $\underline{F_{X_n}(\alpha)} \rightarrow \underline{F_X(\alpha)}$ for each fixed α that is a continuity point of F_X .

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$X_n \xrightarrow{D} X$ if $F_{X_n}(\alpha) \rightarrow F_X(\alpha)$ for each fixed α that is a continuity point of F_X .

"Weak convergence", "convergence in distribution", and
"convergence in law" all mean the same thing.

Summary

Convergence:

- Sure ("pointwise")
- Almost Sure
- In Probability
- Weak ("in distribution"/"in law")

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$$X_n \xrightarrow{D} X \iff F_{X_n}(\alpha) \rightarrow F_X(\alpha) \text{ for each continuity point.}$$

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Stronger
↑

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Stronger



Implies

Break: Convergence of MC integration

$$X_i \sim U([0, 1]^2)$$

(Intuitively)

$$Q_N \equiv \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{X_{i,1}^2 + X_{i,2}^2 \leq 1}(X_i)$$

$$Q_N \rightarrow \frac{\pi}{4} \text{ (sure)?}$$

$$Q_N \xrightarrow{a.s.} \frac{\pi}{4}?$$

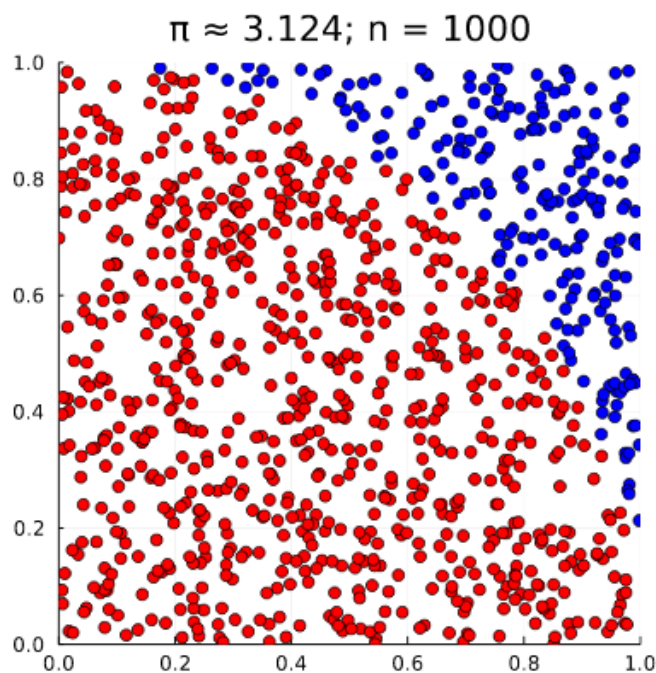
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$$Q_N \xrightarrow{D} \frac{\pi}{4}?$$

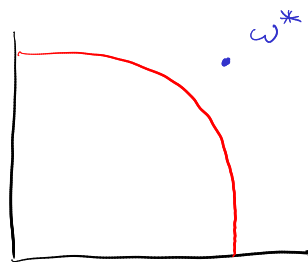
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$$\underline{Q_N} \equiv \frac{1}{N} \sum_{i=1}^N \mathbf{1}_{X_{i,1}^2 + X_{i,2}^2 \leq 1}(X_i)$$



$$X_i(\omega^*) = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



$$Q_N(\omega^*) = 0$$

(Intuitively)

$$Q_N \rightarrow \frac{\pi}{4} \text{ (sure)? } \times$$

$$Q_N \xrightarrow{a.s.} \frac{\pi}{4}?$$



Strong Law
of Large Numbers

$$Q_N \rightarrow_p \frac{\pi}{4}?$$

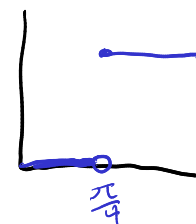


Weak Law
of Large
Numbers

$$Q_N \xrightarrow{D} \frac{\pi}{4}?$$



$$F_{\frac{\pi}{4}}(\alpha)$$



$$F_{Q_N}(\alpha) \rightarrow 0 \quad \forall \alpha < \frac{\pi}{4}$$

$$F_{Q_N}(\alpha) \rightarrow 1 \quad \forall \alpha > \frac{\pi}{4}$$

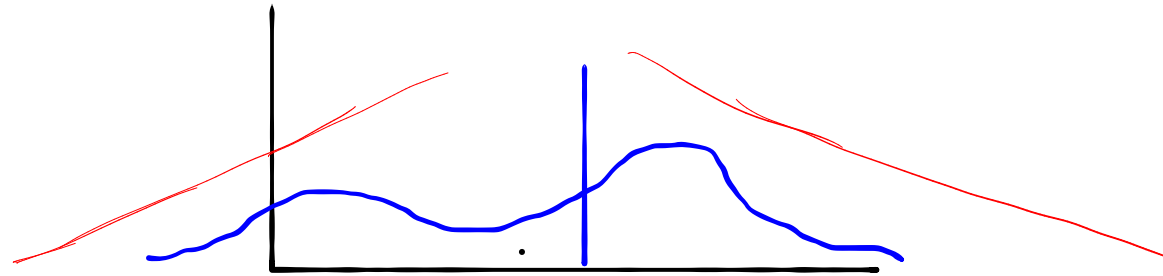
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Intuition: If an r.v. has a finite variance, the probability that a random variable takes a value far from its mean should be small

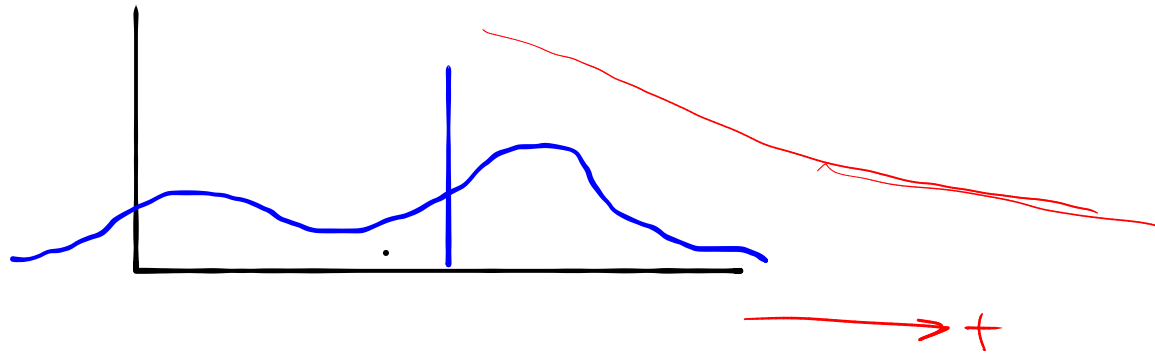
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Concentration inequalities take the form

$$P(X \geq t) \leq \phi(t)$$

where ϕ goes to zero (quickly) as $t \rightarrow \infty$

Concentration Inequalities

Concentration Inequalities

Markov's Inequality:

If $X \geq 0$, then

$$P(X \geq t) \leq \frac{E[X]}{t} \quad \forall t \geq 0$$

Proof:

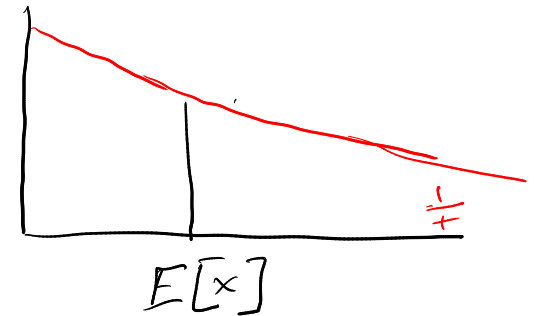
$$E[X] = \int_{-\infty}^{\infty} x p(x) dx$$

$$= \int_0^{\infty} x p(x) dx$$

$$= \int_0^+ x p(x) dx + \int_+^{\infty} x p(x) dx$$

$$\geq \int_+^{\infty} \underbrace{x}_{\geq 0} p(x) dx \geq \int_+^{\infty} + p(x) dx = + \int_+^{\infty} p(x) dx = + P(X \geq +)$$

$$\boxed{\frac{E[X]}{+} \geq P(X \geq +)}$$



Concentration Inequalities

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Let X be *any* real-valued random variable with $\text{Var}(X) < \infty$. Then

Concentration Inequalities

Chebyshev's Inequality:

Let X be *any* real-valued random variable with $\text{Var}(X) < \infty$. Then

$$P(|X - E[X]| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

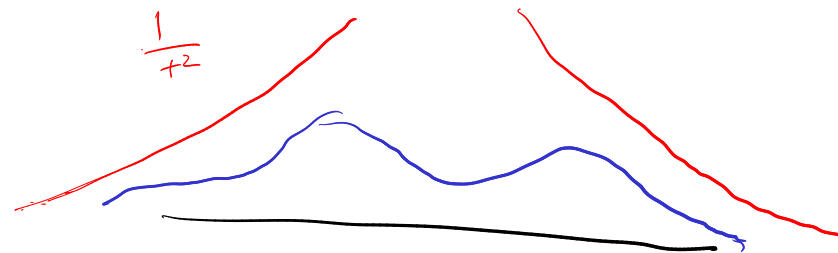
Proof:

$$\text{Var}(X) = E[(X - E[X])^2]$$

Use Markov's inequality on $Y = (X - E[X])^2$

$$P((X - E[X])^2 \geq t^2) \leq \frac{E[(X - E[X])^2]}{t^2}$$

$$P(|X - E[X]| \geq t) \leq \frac{\text{Var}(X)}{t^2}$$



Concentration Inequalities

Chebyshev's Inequality:

Let X be *any* real-valued random variable with $\text{Var}(X) < \infty$. Then

$$P(|X - E[X]| \geq t) \leq \frac{\text{Var}(X)}{t^2}.$$

Very general, but still loose

Concentration Inequalities

Chebyshev's Inequality

k	Min. % within k standard deviations of mean	Max. % beyond k standard deviations from mean
1	0%	100%
$\sqrt{2}$	50%	50%
1.5	55.56%	44.44%
2	75%	25%
$2\sqrt{2}$	87.5%	12.5%
3	88.8889%	11.1111%
4	93.75%	6.25%
5	96%	4%
6	97.2222%	2.7778%
7	97.9592%	2.0408%
8	98.4375%	1.5625%
9	98.7654%	1.2346%
10	99%	1%

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For Normal
Distribution

Concentration Inequalities

Chebyshev's Inequality

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68% For Normal Distribution

95%

99%

Concentration Inequalities

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Moment generating function: $M_X(t) \equiv E[e^{tX}]$

Concentration Inequalities

Moment generating function: $M_X(t) \equiv E[e^{tX}]$

Chernoff Bound: If the moment-generating function M_X exists, then

$$P(X \geq a) \leq \frac{E[e^{tX}]}{e^{ta}} \quad \forall t > 0$$

and

$$P(X \leq a) \leq \frac{E[e^{tX}]}{e^{ta}} \quad \forall t < 0$$

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Proof: Apply Markov's Inequality to e^{+X}

$$P(X \geq a) = P(e^{+X} \geq e^{+a}) \leq \frac{E[e^{+X}]}{e^{+a}}$$

Tighter than Markov and Chebyshev

Bonus Slide: Moment Generating Functions

To get the n th moment from $M_X(t)$, differentiate it n times and set $t = 0$.

Because:

$$\begin{aligned} M_X(t) = \mathbb{E}(e^{tX}) &= 1 + t\mathbb{E}(X) + \frac{t^2 \mathbb{E}(X^2)}{2!} + \frac{t^3 \mathbb{E}(X^3)}{3!} + \cdots + \frac{t^n \mathbb{E}(X^n)}{n!} + \cdots \\ &= 1 + tm_1 + \frac{t^2 m_2}{2!} + \frac{t^3 m_3}{3!} + \cdots + \frac{t^n m_n}{n!} + \cdots, \end{aligned}$$

Concentration Inequalities

Concentration Inequalities

Name	Requirements	Bound

Concentration Inequalities

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Markov	$X \geq 0$, $E[X]$ exists	$P(X \geq t) \leq \frac{E[X]}{t} \quad \forall t \geq 0$

Concentration Inequalities

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Markov	$X \geq 0, E[X] \text{ exists}$	$P(X \geq t) \leq \frac{E[X]}{t} \quad \forall t \geq 0$
Chebyshev	$\text{Var}(X) < \infty$	$P(X - E[X] \geq t) \leq \frac{\text{Var}(X)}{t^2}$

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Exercise

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Let Y be a r.v. that takes values in $[-1, 1]$ with mean -0.5 . Give an upper bound on the probability that $Y \geq 0.5$.

(Weak) Law of large numbers

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Proof:

Exercise: Use Bienaymé + Chebyshev to prove

Law of Large Numbers

Two somewhat astounding takeaways:

Law of Large Numbers

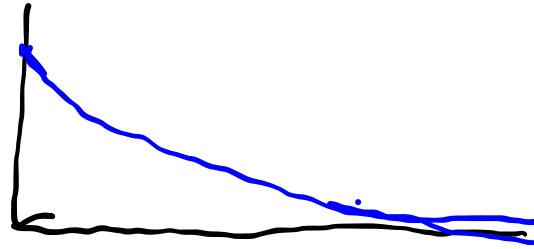
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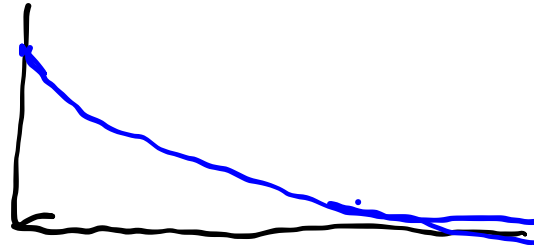
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Two somewhat astounding takeaways:

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2. You can estimate the "standard error" with

$$SE = \frac{s}{\sqrt{N}}$$

where s is the sample standard deviation.